Cambridge Assessment International Education

Cambridge O Level

MATHEMATICS (SYLLABUS D) **4024/21**

Paper 2 **May/June 2022**

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

© UCLES 2022

[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2022

Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

© UCLES 2022

Question	Answer	Marks	Partial Marks
1(a)(i)	2380	**3**	**M1** for $\frac{28}{100}\times 5200$ oe **M1** for $\frac{3}{25}\times 5200$ oe
1(a)(ii)	751.1[0]	**2**	**M1** for $740 + \frac{1.5}{100}\times 740$ oe or **B1** for 11.1[0]
1(b)(i)	85.14	**1**	
1(b)(ii)	1400	**3**	**M1** for $\frac{215.60}{2.20}$ soi **M1** for $\frac{their 98}{7}\times 100$ oe
1(b)(iii)	1.82 or 1.818…	**2**	**M1** for $\frac{2.24-2.20}{2.20}[\times 100]$ oe or $\frac{2.24}{2.20}\times 100$ oe
2(a)	18 22 32 50 50 72	**2**	**B1** for one row or column correct
2(b)	$4n + 2$ oe final answer	**2**	**B1** for $4n + k$ oe seen
2(c)	24	**2**	**M1** for *their* $(4k + 2) = 98$
2(d)	$2n^2$ oe final answer	**2**	**B1** for answer $kn^2 + \ldots$, $k \neq 0$ or second difference = 4 soi
2(e)	882	**2**	**M1** for $2\times 20^2 + 4\times 20 + 2$ oe or $n = 20$ substituted into *their*(b) + *their*(d) oe
3(a)(i)	27.8[0] to 28.4[0]	**1**	
3(a)(ii)	2.45 to 2.55	**2**	**B1** for 18.5 soi
3(b)	4.4	**2**	**M1** for $\frac{10}{6+8+11}[\times 11]$ or $\frac{11}{6+8+11}[\times 10]$
3(c)	3.7 nfww	**3**	**M2** for $8.8125 = 2.35(d + 0.05)$ oe or **B2** for answer 3.75 OR **B1** for 2.35 seen **M1** for $\frac{8.8125}{their 2.35}$ or for $\frac{8.8125 - their 2.35\times 0.05}{their 2.35}$ oe

Question	Answer	Marks	Partial Marks
4(a)	2.5 or $2\frac{1}{2}$	**2**	**M1** for $x \times x \times 10 = 62.5$ oe
4(b)(i)	$\frac{84}{360} \times \pi \times 2 \times 15$ leading to 7π or $\frac{84}{180} \times \pi \times 15$ leading to 7π	**1**	
4(b)(ii)	3.5 or $3\frac{1}{2}$	**2**	**M1** for $2 \times \pi \times r = 7\pi$ oe or $\frac{84}{360} \times \pi \times 15^2 = \pi \times r \times 15$ oe
4(b)(iii)	14.6 or 14.58 to 14.59	**2**	**M1** for $15^2 - (their 3.5)^2$ oe
4(c)	18 [minutes] 17 [seconds] cao	**3**	**M2** for $\frac{\pi \times 20^2 \times 80}{5500}$ or **M1** for $\pi \times 20^2 \times 80$ or $\frac{their\text{volume}}{5500}$
5(a)	–12	**1**	
5(b)	Correct smooth curve	**3**	**B2FT** for 6 or 7 points correctly plotted or **B1FT** for 4 or 5 points correctly plotted
5(c)	Ruled line $y = 5$	**M1**	
	–1.8 to –1.5 –0.6 to –0.3 2.1 to 2.5	**A2**	**A1** for two correct After 0 scored, **SC1** for three correct solutions or $x^3 - 4x + 3 = 5$ soi or for three correct solutions from their horizontal line drawn
6(a)	44 to 48 with correct ruled line drawn	**2**	**B1** for correct ruled line drawn or *their* angle measured from a ruled line from D to AB.
6(b)	Arc 6.5 cm centre B, angle bisector of angle BCD and correct region shaded	**4**	**FT** *their* path from D to AB **B1** for angle bisector of angle BCD **B2** for arc 6.5 cm centre B or **B1** for arc centre B
6(c)	130 to 150 nfww	**1**	
7(a)(i)	[0].22 oe	**1**	
7(a)(ii)	[0].208 oe	**2**	**M1** for 0.26×0.4 [× 2] oe

Question	Answer	Marks	Partial Marks
7(a)(iii)	[0].64 oe	**3**	**M2** for 1 – 0.6×0.6 oe or 0.4×0.4 + 0.4×0.6×2 oe or **M1** for 0.6×0.6 oe seen or 0.4×0.6[×2] oe seen or 0.4×0.4 oe seen
7(b)(i)	Correct histogram drawn	**3**	**B1** for 3 more rectangles on correct bases **B1** for 3 more correct frequency densities soi
7(b)(ii)	22	**2**	**M1** for 0.5×*their*24 +10 oe or $10 + \frac{2}{3}(18)$ oe
8(a)(i)	$5x + 10y = 40$ leading to $x + 2y = 8$	**1**	
8(a)(ii)	$4x + 3y = 19$	**B1**	
	Correct method to eliminate one variable	**M1**	
	$x = 2.8[0]$ $y = 2.6[0]$ cao	**A2**	**A1** for either $x = 2.8[0]$ or $y = 2.6[0]$ nfww After A0 scored, **SC1** for a pair of values that satisfy either correct equation or for correct answers with no working or with incorrect
8(b)	$1 < x < 4\frac{3}{4}$ oe final answer	**3**	**B2** for $1 < x$ or $x < 4\frac{3}{4}$ or **M1** for $-\frac{8}{4}$ […] x – 3 […] $\frac{7}{4}$ or better or –8 + 12 […] 4x […] 7 + 12 or better
8(c)	$2x^2 - 9x - 13$ [= 0]	**B3**	**B2** for $\frac{4(2x+3)+2(x-1)}{(2x+3)(x-1)}$ [=1] or better or **B1** for 4(2x + 3) + 2(x – 1) oe or for (2x + 3)(x – 1) or better
	$\frac{-(-9) \pm \sqrt{(-9)^2 - 4\times2\times-13}}{2\times2}$ oe or $\frac{-(-9)}{2\times2} \pm \sqrt{\left(\frac{(-9)}{2\times2}\right)^2 - \left(\frac{-13}{2}\right)}$	**B2**	**FT** *their* 3-term quadratic **B1FT** for $\sqrt{(-9)^2 - 4\times2\times-13}$ oe or for $\frac{-(-9) \pm \sqrt{\textit{their}\ \text{discriminant}}}{2\times2}$ or for $\left(x + \frac{-9}{2\times2}\right)^2$
	5.65 and –1.15	**B1**	

Question	Answer	Marks	Partial Marks
9(a)	330°	**1**	
9(b)	365 or 365.3…	**4**	**B1** for $\left[B\hat{A}C=\right]$ 78 **M2** for $\sqrt{300^2+280^2-2\times300\times280\cos \textit{their}78}$ or **B2** for 133000 or 133400 to 133500 or **M1** for $300^2+280^2-2\times300\times280\cos \textit{their}78$
9(c)	[0]47.3 or [0]47.25 to [0]47.26 and [0]96.7 or [0]96.74 to [0]96.75	**5**	**B4** for [0]47 and [0]97 nfww or [0]47.3 or [0]47.25 to [0]47.26 or [0]96.7 or [0]96.74 to [0]96.75 or **B3** for $\left[D\hat{A}B=\right]$24.7 or 24.74 to 24.75 or **M2** for $\sin A=\frac{145\sin120}{300}$ or **M1** for $\frac{145}{\sin A}=\frac{300}{\sin120}$ After 0 scored **SC1** for sketch with 120 and 145 in correct place or 72 + *their DAB* or 72 – *their DAB*
10(a)	Translation $\begin{pmatrix}-4\\3\end{pmatrix}$	**2**	**B1** for translation or $\begin{pmatrix}-4\\3\end{pmatrix}$ with no further properties
10(b)	$\begin{pmatrix}-1&0\\0&1\end{pmatrix}$	**1**	
10(c)	Triangle at (–1, 0) (–1, –3) (5, –3)	**2**	**B1** for correct size and orientation, wrong centre After 0 scored, **SC1** for enlargement with correct centre but wrong scale factor
11(a)	8.25 or 8.246…	**2**	**M1** for $(3-1)^2+(-3-5)^2$ oe
11(b)	$y=\frac{1}{4}x+\frac{1}{2}$ oe	**5**	**B1** for midpoint (2, 1) **M1** for [grad PQ =] $\frac{-3-5}{3-1}$ oe soi **M1** for [grad perpendicular =] $\frac{-1}{\textit{their}(-4)}$ **M1** for substitution of *their* (2, 1) into $y=(\textit{their}\frac{1}{4})x+c$ oe

© UCLES 2022